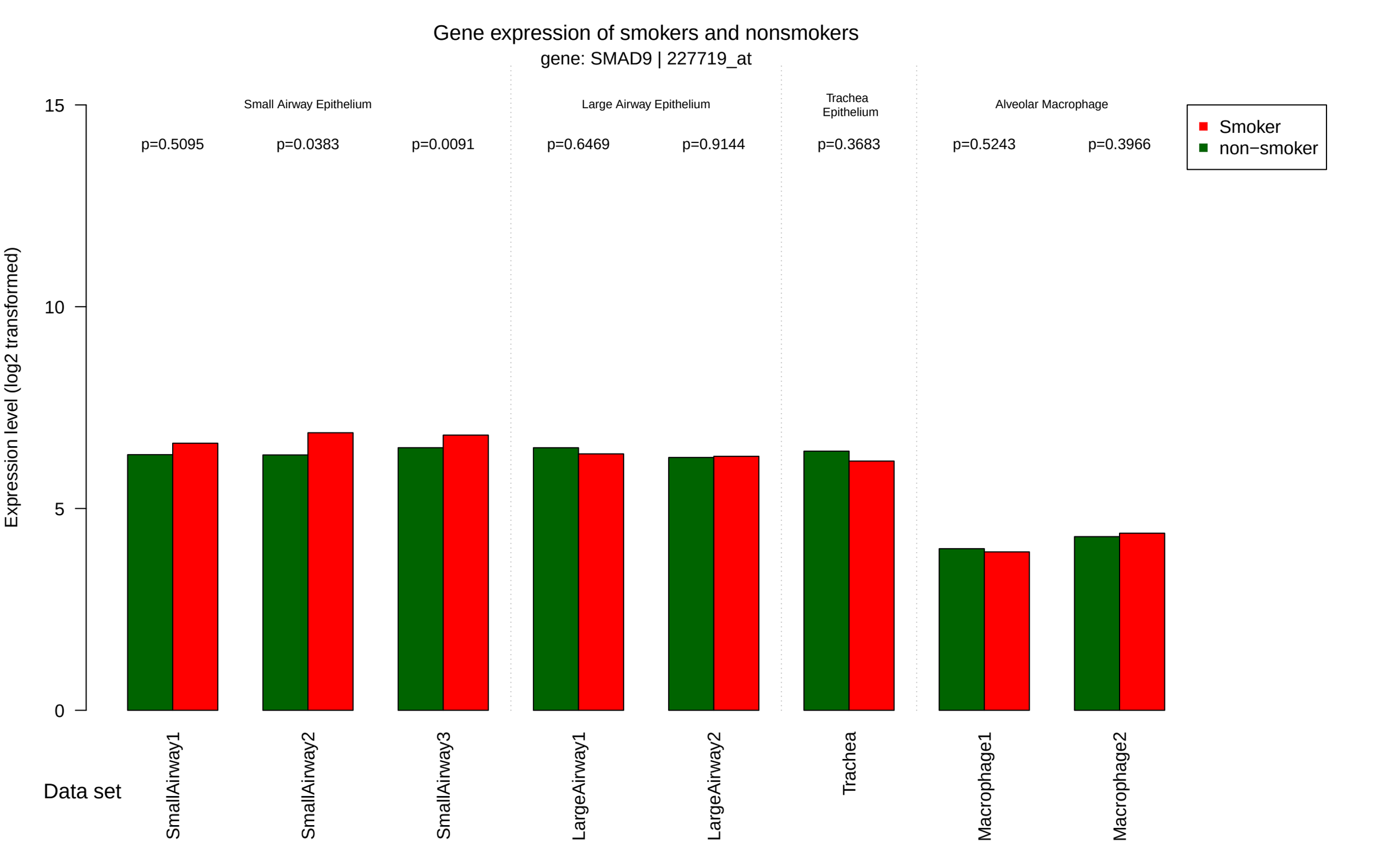
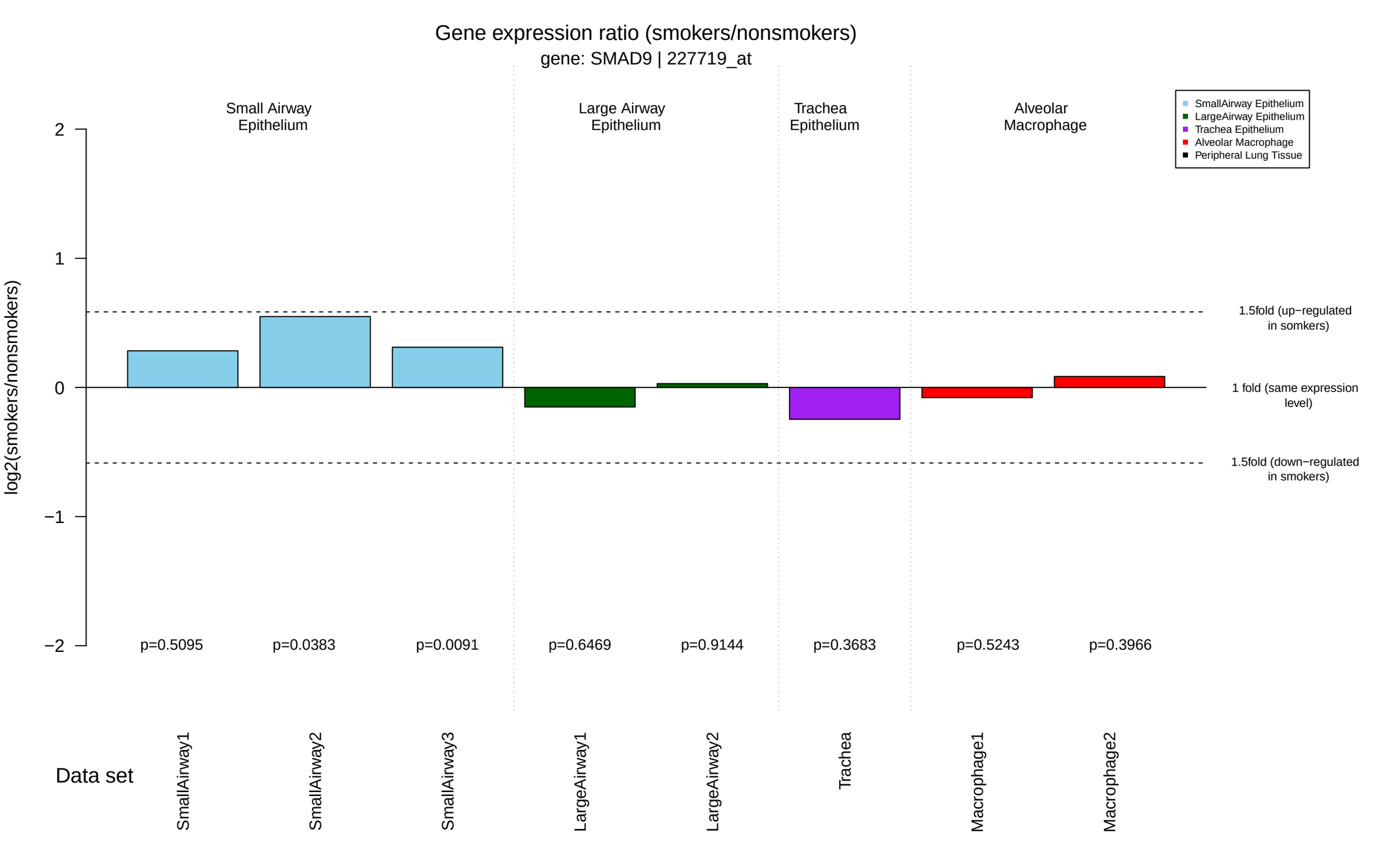
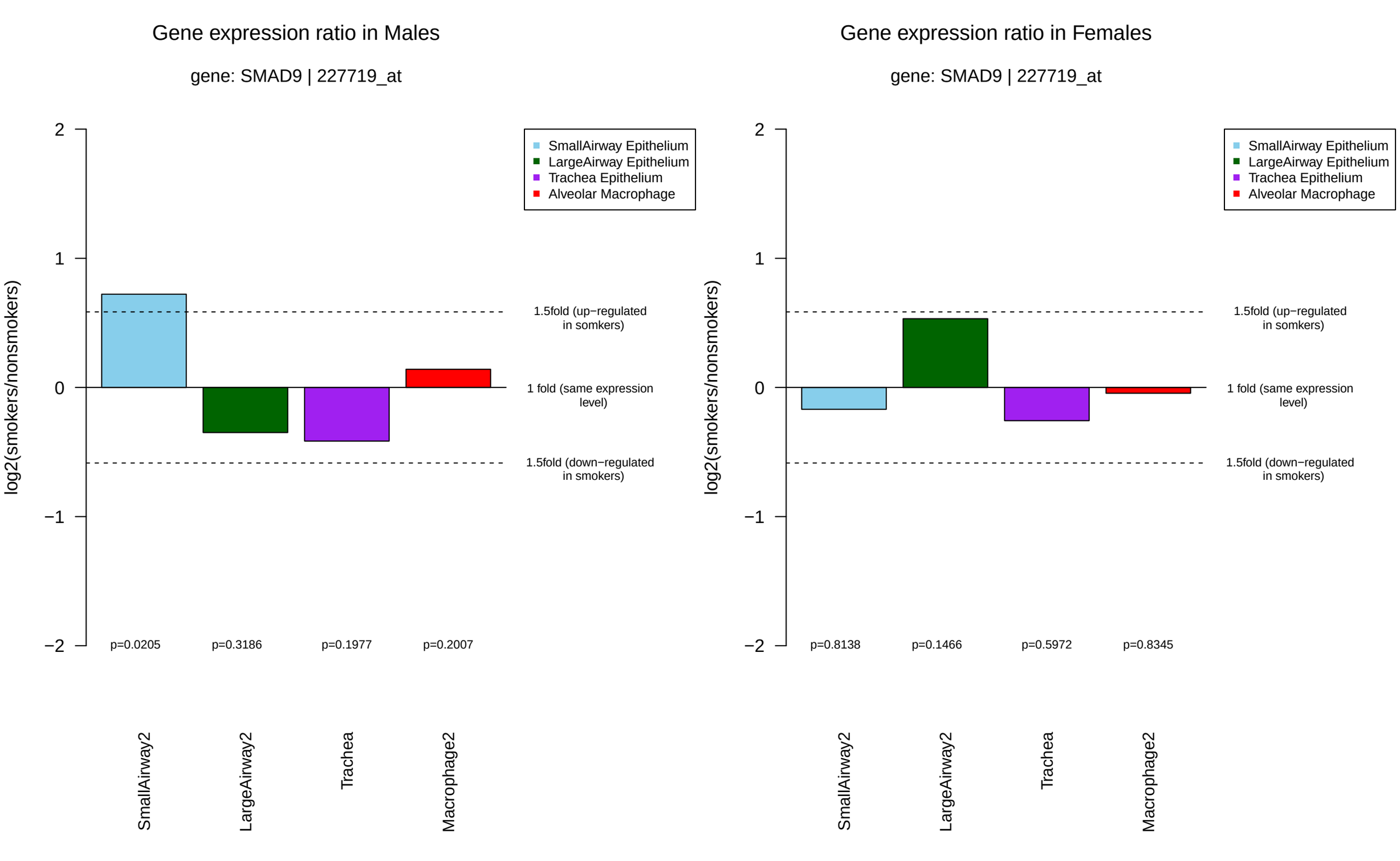
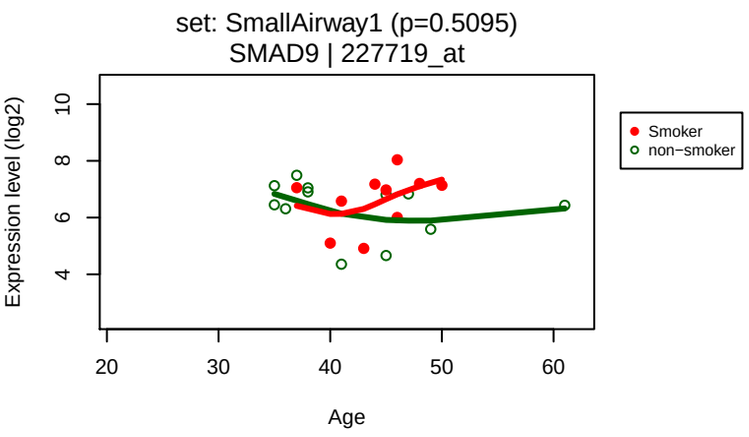
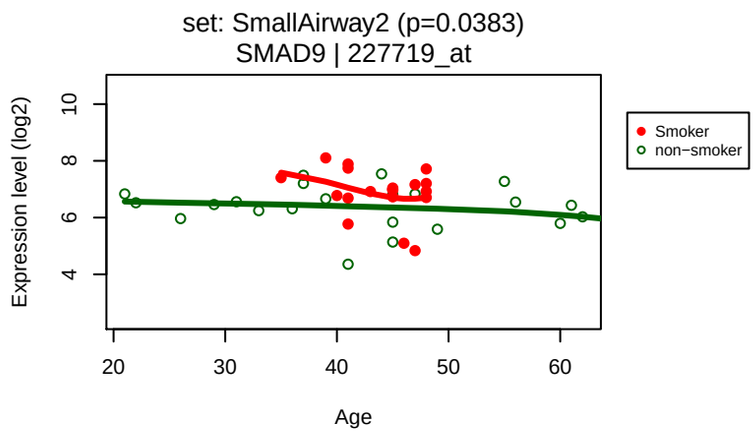
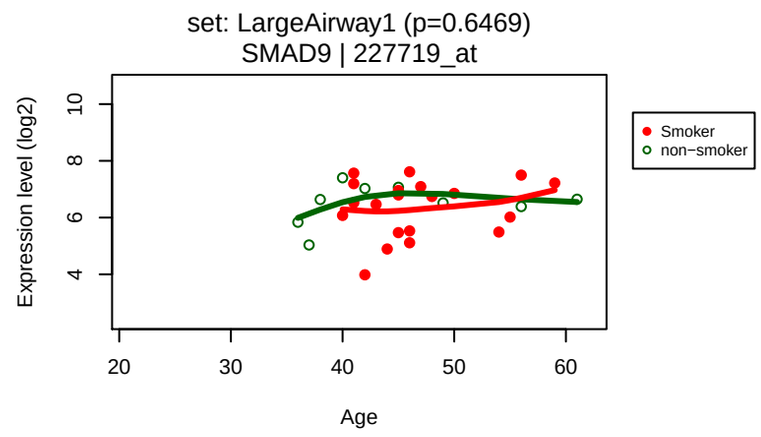

Gene expression of smokers and nonsmokers
gene: SMAD9 | 227719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5095
p=0.0383
p=0.0091
p=0.6469
p=0.9144
p=0.3683
p=0.5243
p=0.3966
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

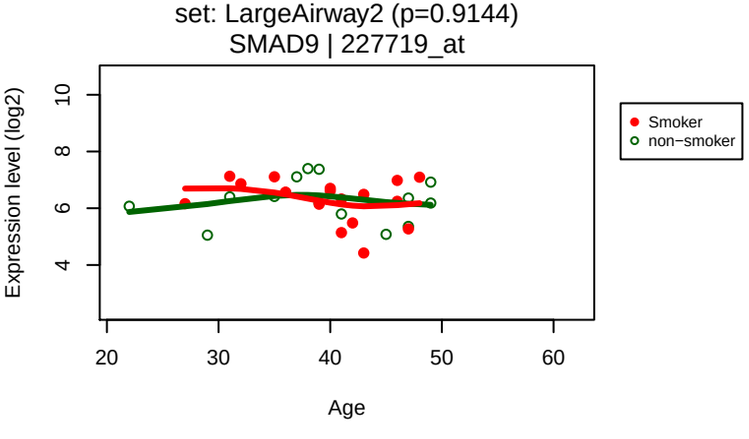

Gene expression ratio (smokers/nonsmokers)
gene: SMAD9 | 227719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5095
p=0.0383
p=0.0091
p=0.6469
p=0.9144
p=0.3683
p=0.5243
p=0.3966
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

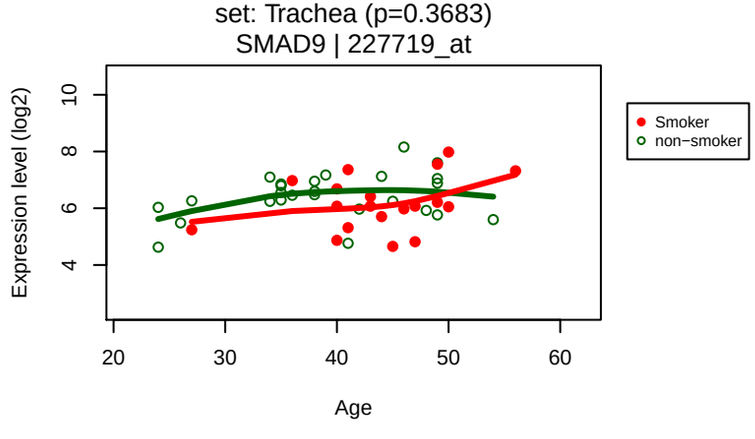

Gene expression ratio in Males
gene: SMAD9 | 227719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0205
p=0.3186
p=0.1977
p=0.2007
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SMAD9 | 227719_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8138
p=0.1466
p=0.5972
p=0.8345
SmallAirway2
LargeAirway2
Trachea
Macrophage2

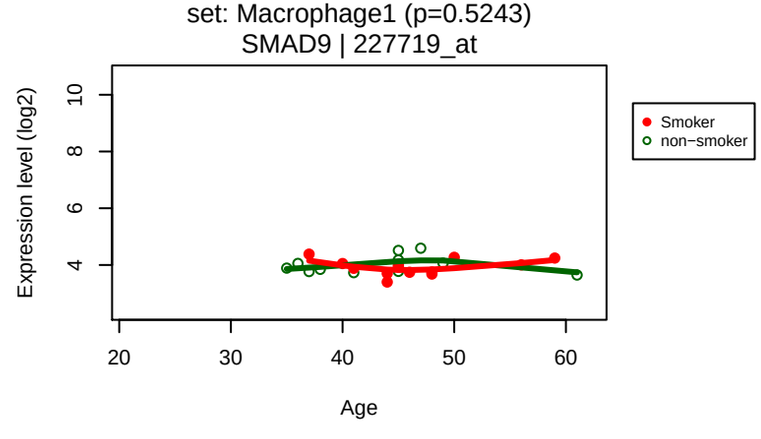

set: SmallAirway1 (p=0.5095)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker

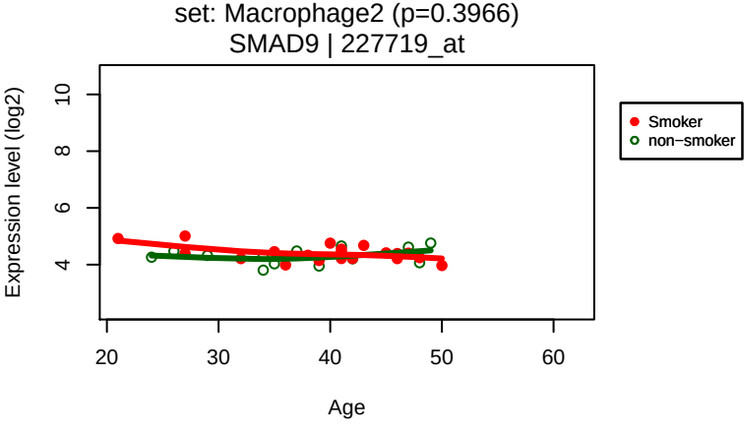

set: SmallAirway2 (p=0.0383)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker

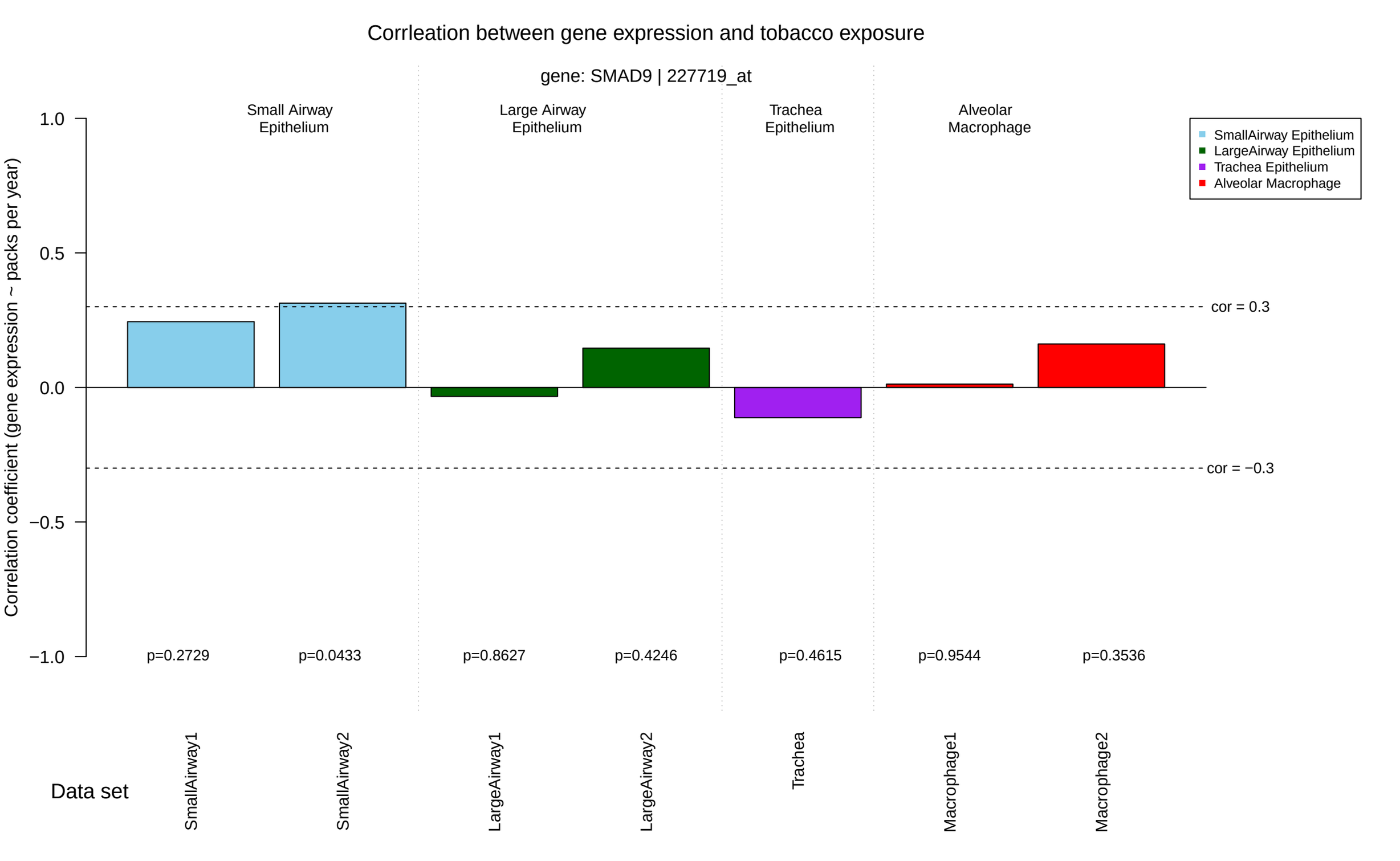

set: LargeAirway1 (p=0.6469)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9144)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3683)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5243)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3966)
SMAD9 | 227719_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SMAD9 | 227719_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2729
p=0.0433
p=0.8627
p=0.4246
p=0.4615
p=0.9544
p=0.3536
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2